Ein Mishpat as Training Data: Contrastive Retrieval over the Babylonian Talmud

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Abstract

We observe that the *ein mishpat / ner mitsvah* apparatus — a cross-reference system printed in the margins of the Vilna Talmud, mapping passages of Talmudic argument to the medieval legal codes that rule from them — constitutes a large, hand-curated, cross-register relevance dataset assembled centuries before information retrieval existed. We extract **50,214** such pairs from the Sefaria link graph, of which **27,013** connect Maimonides' *Mishneh Torah* to the Babylonian Talmud at segment granularity with English on both sides, and use them to fine-tune a retrieval model.

Measurement precedes training. We establish that lexical and dense retrieval score almost identically on code-to-source retrieval (MRR 0.497 vs 0.500) while succeeding on *different* queries (top-10 overlap 63%, union R@10 0.836); that agent-derived semantic chunking into *sugyot* produces a large apparent gain which *reverses sign* under a candidate-pool control; and — decisively — that retrieval measured with code rulings as queries does not transfer to the natural questions a user actually asks. That last finding redirects the training data: anchors must be question-shaped, not ruling-shaped.

The resulting model, **torah-embed**, is fine-tuned from `bge-base-en-v1.5` on 33,393 triplets with same-daf hard negatives, and evaluated on 552 queries from two entirely held-out tractates against the full 81,481-segment corpus. It improves MRR by +12.2% on ruling queries, +20.0% on practical questions and +**35.9**% on conceptual questions — a gain that rises monotonically as lexical overlap with the target falls (0.41 → 0.30 → 0.21). Against a pre-registered criterion requiring that question-query gains exceed ruling-query gains, the model passes under both credit rules (+27.9% vs +12.2% strict; +28.9% vs +14.2% sugya-level). Absolute performance remains modest, and we say so: conceptual questions reach MRR 0.116.

We report the measurements, the four designs they eliminated, and the five errors we made obtaining them — three of which inflated results and were caught only by controls fixed in advance.

1 Introduction

Retrieval over the Talmud is an unusually well-posed problem with an unusually awkward corpus. The Babylonian Talmud is roughly 1.8 million words of Aramaic and Hebrew dialectic, organised not by topic but by association; a reader looking for the source of a legal ruling must know where to look, and knowing where to look is most of the skill.

The motivating application is a retrieval-augmented system that, given a question, returns the passage of Talmudic argument bearing on it. Such a system needs training data: pairs of (query, relevant passage) at sufficient volume to fine-tune an embedding model. Prior work of ours [1] obtained such pairs for mathematical retrieval by extracting a knowledge graph from the corpus with a language model and deriving (concept, chunk) pairs from it. That supervision

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was synthetic, approximate — a concept mapped to *every* chunk of every relevant paper — and its coarseness forced reliance on MRR and NDCG because Recall@20 sat at 3–4%.

The Talmud offers something better, and it is the observation this paper is built around.

1.1 *Ein mishpat* is already a relevance dataset

Printed in the margin of every page of the Vilna Talmud is a cross-reference apparatus, *ein mishpat / ner mitsvah*, compiled in the sixteenth century by Joshua Boaz. It maps each passage of Talmudic argument to the places in the medieval legal codes — Maimonides' *Mishneh Torah*, the *Tur*, the *Shulchan Arukh* — that rule from it. Sefaria has digitised this apparatus as typed links at segment granularity.

This is, in modern terms, a relevance dataset with properties one would struggle to buy:

- **Hand-curated** by domain experts, over centuries, with no model in the loop.
- **Cross-register by construction.** One side is terse codified law; the other is discursive dialectic. They share little surface vocabulary, so the pairs test semantic rather than lexical matching.
- **Segment-level, not document-level.** Relevant sets are small — mean 2.6 targets per source — so Recall@k is a sharp metric rather than the blunt instrument it became in [1].
- **Exactly the deployment task.** "Where does this ruling come from" is the question the apparatus was built to answer.

The analogy to SPECTER2 [2] is close and, we think, load-bearing: SPECTER2 improves a scientific encoder by contrastive training on citation triplets. *Ein mishpat* is a citation graph. What follows is an attempt to establish, by measurement, whether that analogy survives contact with the data.

1.2 Contributions

This is a measurement paper and its contributions are measurements.

1. A census of the available supervision: 50,214 cross-register pairs, and a coverage analysis showing which parts of the canon are usable in English at all (§2).
2. Baselines showing that lexical and dense retrieval are equal in aggregate and complementary in detail (§3).
3. A negative result: agent-derived semantic chunking does not improve retrieval, and the apparent improvement inverts under a proper control (§4).
4. A query-distribution result: the benchmark task is not the deployed task, and the gap is large but partly an artifact of metric granularity (§5).
5. A methodological account, including five errors we made and how each was caught (§6).

2 Corpus and supervision census

2.1 Coverage determines scope

All texts were obtained from the Sefaria API. The corpus was fixed as English translation; this rules out rabbinic-Hebrew encoders such as BEREL [4] and admits the mature English retrieval-model ecosystem, at the cost of depending on translation coverage.

Coverage is extremely uneven. Table 1 reports the fraction of Hebrew segments having an English counterpart, measured per segment.

Text	English coverage	Consequence
Talmud Bavli (Davidson)	100%	corpus
Mishneh Torah (Touger)	97.9%	**gold training set**
Mishnah	93.9%	positive control only (see below)
Tosafot	27%	dropped
Rashi on the Talmud	9.5%	dropped
Shulchan Arukh	5.4%	dropped

Table 1: English coverage by source, measured at segment granularity over the Sefaria API. An earlier estimate of 89% for *Shulchan Arukh* was obtained by sampling chapters 1, 50 and 150 of each book; early chapters are translated and the long tail is not.

The consequence is severe and clarifying. Rashi and *Tosafot* would have supplied millions of pairs and are unusable. But those pairs carried a hazard: a Rashi comment opens with a *dibbur hamatchil*, a lemma copied verbatim from the Talmudic text, so training on them risks learning substring matching rather than semantics. Choosing English deletes the bulk of the supervision and the dominant leakage channel together.

2.2 The census

Links were harvested from the Sefaria API and filtered to the 37 tractates of the Bavli proper.

Source	Unique pairs	Distinct sources	Distinct Bavli targets
Mishneh Torah → Bavli	27,601	9,675	21,300
Shulchan Arukh → Bavli	15,860	6,968	12,494
Mishnah → Bavli	6,753	2,932	6,056
Combined	**50,214**	—	27,573

Table 2: Cross-register link census. All 37 tractates are covered; 79 of 88 *Mishneh Torah* books contribute. The figure is a floor: *Shulchan Arukh, Even HaEzer* is absent entirely, its index using a schema node for which our chapter enumeration returned nothing.

Of the *Mishneh Torah* pairs, 25,448 carry the link type *ein mishpat / ner mitsvah* — that is, the supervision is dominated by the classical apparatus rather than by modern editorial additions.

2.3 Two pair types are unusable, for opposite reasons

Shulchan Arukh pairs are numerous but only 855 of 15,860 have English on the source side. They are dropped.

Mishnah pairs are well covered but **lexically duplicate**: the Gemara quotes each mishnah verbatim at the head of its sugya and the Davidson edition reproduces the quotation, so source and target are frequently the same English sentences. Mean token overlap is 0.62, with 34/158 sampled pairs above 0.9. Training on them teaches copy detection.

They are nonetheless retained, **as a positive control**: a correctly wired lexical retriever must score near ceiling on duplicate pairs. It does (§3), which is how we know the harness is sound.

Mishneh Torah pairs are clean by the same measure: mean token overlap 0.42, and none of 171 sampled pairs above 0.9. This is the gold set: **27,013** pairs over ≈ 9,500 distinct queries with English on both sides — two orders of magnitude more supervision than [1] was trained on, and human rather than model derived.

3 Baselines

3.1 Setup

The corpus is the full English Bavli: **81,481** segments, mean 99 words, median 96, 90th percentile 149. Segments are short enough that no chunking is required for a 512-token encoder.

Queries are source-side texts; relevance is the link. Retrieval is over the entire corpus with no tractate hint or candidate pruning. We report BM25 ($k_1 = 1.5$, $b = 0.75$) and `bge-base-en-v1.5` [3] used zero-shot, with the prescribed query instruction prefix.

3.2 Results

	BM25			`bge-base` zero-shot		
Split	MRR	R@1	R@10	MRR	R@1	R@10
Mishneh Torah (n=171)	0.497	0.380	0.731	0.500	0.374	0.737
Mishnah, control (n=158)	0.740	0.684	0.892	0.684	0.601	0.842
Shulchan Arukh (n=171)	0.358	0.269	0.550	0.416	0.327	0.596

Table 3: Zero-shot baselines over 81,481 segments. The Mishnah row is a control: BM25 approaching ceiling on lexically duplicate pairs confirms the harness is correct.

Two observations. First, the task is hard and unsolved: neither method exceeds MRR 0.5 on the gold split. Second, and less expected, dense retrieval offers *no aggregate advantage* over BM25 on cross-register retrieval, despite the pairs having been selected precisely because they share little vocabulary.

3.3 The aggregate hides the finding

Scoring 0.497 and 0.500, the two methods look interchangeable. They are not.

Mishneh Torah, n=171	R@10
BM25 alone	73.1%
`bge-base` alone	73.7%
Both	63.2%
Union (either)	**83.6%**
Neither	16.4%

Table 4: Complementarity at rank 10. Found by BM25 only: 17 queries. By dense only: 18. The symmetry is striking.

Ten points of recall are available in the union that neither method reaches alone, and an unweighted z-score fusion realises part of it with no training at all (MRR 0.497/0.500 $\rightarrow$ 0.527).

We record this as a methodological warning as much as a result. A single mean over a heterogeneous query set supported the conclusion "dense adds nothing", which is false. The useful question about any retriever here is not whether the average moved but *which queries it fixes*.

4 Semantic chunking: a negative result

4.1 Motivation and protocol

A Sefaria segment averages 99 words and an *amud* holds 15.2 of them, while a *sugya* — a complete unit of argument — spans five to fifteen. Every indexed unit is therefore a fragment of an argument. We tested whether merging segments into sugyot improves retrieval.

Boundaries were chosen by language-model agents that read the text. Nine agents processed 32 amudim of tractate Berakhot under a protocol designed for verifiability:

- Agents return **reference ranges, never text**, making corruption of the corpus structurally impossible.
- Boundaries fall only at existing segment edges; since Sefaria segments never break mid-sentence, mid-sentence cuts cannot occur.
- Output must **partition** the input: every segment in exactly one chunk. This is checked mechanically.
- Each chunk carries a generated heading giving its context in the surrounding argument, prepended to the body before embedding.
- Headings may not name any later legal code, since the evaluation measures exactly the link to those codes.

The protocol produced 216 chunks with **zero partition failures** and zero heading leaks. Merge ratio was 26.4 segments to 6.8 chunks per amud.

4.2 The control matters more than the result

Chunking reduces the candidate pool (844 segments → 209 chunks in the evaluated region). A smaller pool mechanically improves rank-based metrics. We therefore report both raw metrics and **matched-fraction recall**: the top 1% of the pool in each condition (8 segments vs 2 chunks), which also approximately matches retrieved text volume.

Condition	Method	Pool	MRR	R@10	**R@1%**
A. Segments	BM25	844	0.580	0.788	**0.747**
A. Segments	dense	844	0.527	0.818	**0.818**
B. Sugya chunks	BM25	209	0.683	0.919	0.677
B. Sugya chunks	dense	209	0.587	0.838	0.596
C. Chunks + heading	BM25	209	0.681	0.919	0.687
C. Chunks + heading	dense	209	0.595	0.838	**0.646**
D. Heading only	dense	209	0.415	0.707	0.384

Table 5: Chunking A/B, *Mishneh Torah* queries, n=99, Berakhot 2a–17b. Raw MRR and R@10 favour chunking; matched-fraction recall reverses the verdict.

Read raw, chunking is a triumph: MRR 0.580 → 0.683, R@10 0.788 → 0.919. Under the matched-fraction control it is a *loss*: dense R@1% falls 0.818 → 0.646.

The mechanism is dilution. A chunk containing the gold segment also contains three or four others, and the embedding averages toward their mean. This is the precision cost of context, and it is real.

We emphasise that the confound did not merely inflate the effect; **it reversed its sign**. Reported without the control, this experiment would have shown a ten-point gain and written semantic chunking into the system.

4.3 Two results worth keeping

Conditions B and C share a pool, so their comparison is unconfounded. Headings help dense retrieval modestly and at no cost (R@1% 0.596 → 0.646) and do nothing for BM25, which cannot exploit context it can only match lexically.

Condition D is a **behavioural leakage detector**. If the agents had encoded answers in the headings, heading-only retrieval would be excellent. It scores MRR 0.415 against 0.595 for heading-plus-body — substantially worse than the text itself. This is stronger evidence than a string check, and we recommend retaining such a condition whenever generated context is added to an index.

4.4 Resolution: retrieve narrow, present wide

The metric assumes segment-level precision, because *ein mishpat* links point at segments. Semantic chunking disputes that premise, arguing the natural unit is the argument. The evaluation therefore encodes the assumption the design contests.

We resolve this by separating the two roles. The **segment** remains the indexing and retrieval unit; the **sugya** becomes the display and context unit. The system retrieves a segment and returns the enclosing argument. This takes the precision of fine granularity and the readability of coarse, and uses the chunking work rather than discarding it.

5 The benchmark task is not the deployed task

5.1 A paired test

Every measurement above uses a *Mishneh Torah* ruling as the query. A deployed system receives a question. To isolate phrasing from target, we took 40 rulings with known gold sugyot and had agents rewrite each as the question a learner would ask, in plain English, avoiding the ruling's technical vocabulary, then scored retrieval against *the same* gold.

Credit	Method	ruling	question	change
strict	BM25 (MRR)	0.514	0.145	−72%
strict	dense (MRR)	0.451	0.271	−40%
strict	BM25 (R@10)	0.750	0.325	−57%
strict	dense (R@10)	0.725	0.500	−31%
sugya	BM25 (R@10)	0.800	0.375	−53%
sugya	**dense (R@10)**	0.750	**0.675**	**−10%**
sugya	dense (MRR)	0.607	0.346	−43%

Table 6: Paired comparison, n=40, same gold targets. "Sugya" credit accepts a retrieved segment within ±3 of a gold segment, approximating the unit the system displays.

5.2 What survives

Under strict scoring the collapse looks total. It is not, and the difference matters for what to build.

BM25 genuinely collapses, on both metrics under both credit rules. Its advantage in Table 3 was an artifact of using rulings as queries: a ruling shares vocabulary with its source in a way a question does not.

Dense retrieval largely survives. Scored at the granularity the system displays, its recall falls only 10%. What degrades is ranking: MRR falls 43%. Dense retrieval finds the right argument for a natural question nearly as often as for a ruling, and cannot put it first.

Notably, token overlap with the gold passage is unchanged between query forms (0.41 vs 0.42). The collapse is not explained by vocabulary loss but by *specificity*: a question is short and its words are common, so it carries few discriminative terms.

5.3 A third query distribution

We also ran 26 questions drawn from the author's own writing — theological and historical rather than practical (*Why did Jacob change his name to Israel?*; *Is the promise of the land conditional on how we behave?*). No gold labels exist for these; results were inspected qualitatively and are reproduced in full in `questions.pdf`.

Retrieval was substantially better than the benchmark predicted. Two examples: *Why did Jacob change his name to Israel?* returns Berakhot 13a:1, the sugya on precisely that question; *Who counts as the firstborn?* returns Bekhorot 47b:14 on firstborn status for inheritance.

Top-1 agreement between BM25 and dense was **2 of 26**, against 63% on the synthetic benchmark — complementarity is far stronger on out-of-distribution queries. On the Akedah question the two retrievers returned the two halves of a two-part question: BM25 the sacrifice narrative, dense the promise of descendants "as the stars". A fused ranking would have buried one of them.

6 Method, and five errors

We record our errors because each was caught by the same signal — a number that flattered the project — and because the practices that caught them are the transferable part of this work.

1. **A self-fulfilling denominator.** Coverage was computed as non-empty over segments-*returned*; texts with no English version returned nothing and left the denominator. The metric could only ever report 100%.
2. **An over-broad filter.** Counting every title under `Talmud > Bavli` as Talmud included commentaries *on* the Talmud, inflating the census from 27,601 to 39,204 pairs. It was noticed because a commentary appeared in a ranking of tractates.
3. **A systematic constraint.** Requiring chunks to partition within an *amud* forced every sugya crossing a page break to be severed — the exact failure the exercise existed to prevent, occurring at all 5,349 boundaries.
4. **An unrepresentative sample.** *Shulchan Arukh* coverage sampled chapters 1, 50 and 150; the estimate was 89% and the truth 5.4%.
5. **A sign-reversing confound.** The chunking A/B without pool-size control (§4).

Four practices proved to carry their weight:

- **Pre-registration.** Acceptance criteria were written before results existed — the chunking verifier before the agents returned, the decision rule before the dense baseline, the matched-fraction control before the A/B.
- **Adversarial controls in every table.** The BM25 row as a lexical-leakage alarm; the Mishnah positive control; heading-only retrieval as a leakage detector.
- **Structural rather than procedural guarantees.** Agents returning ranges rather than text cannot corrupt a corpus, whatever they do.

- **Asking agents what they were unsure about.** Errors 3 above, and a prompt defect in which a stated "typical" chunk count became a target that fragmented arguments, were *invisible* to every mechanical check: a size-motivated split and a severed sugya both partition perfectly. Both surfaced only through agents volunteering doubt about work that verified clean. An A/B on the corrected prompt confirmed the effect: 23% fewer chunks, never more on any amud.

7 The model

7.1 Setup

Base: `bge-base-en-v1.5` (110M parameters) — the strongest English retriever at BERT-base scale that fine-tunes within 16 GB of unified memory. **Loss:** MultipleNegativesRankingLoss. **Anchors:** both ruling-shaped and question-shaped, the latter generated by language-model agents in two registers (practical and conceptual) and oversampled 3× to offset their smaller number. **Negatives:** up to three per query drawn from the *same daf*, non-linked — random in-batch negatives are trivially easy in this corpus, since a random segment is another tractate on another subject.

Training: 33,393 triplets, 1 epoch, batch 16, max length 192, lr 2×10^{-5} with 10% warmup, 2,087 steps in 1h25m on Apple MPS. Loss fell 0.641 → 0.303. Batch size was forced by memory; the method prefers 64 or more, and this is the most likely source of further gain.

7.2 Evaluation protocol, fixed before training

Splitting is by **whole tractate** — Sanhedrin and Yevamot held out entirely — because splitting by link leaks: both endpoints of a held-out link still appear via their other links. The 809 queries with targets on both sides were *discarded* rather than assigned. Retrieval is over all 81,481 segments with no pruning and no tractate hint; pruning to the held-out tractates would shrink the pool 20× and inflate every metric, which is precisely the confound that reversed the chunking result in §4.

7.3 Results

Query form	BM25	zero-shot	**torah-embed**	gain
ruling	0.430	0.412	**0.462**	+12.2%
practical question	0.090	0.175	**0.211**	+20.0%
conceptual question	0.042	0.086	**0.116**	**+35.9%**
ruling, sugya credit	0.509	0.491	**0.561**	+14.2%
practical, sugya credit	0.125	0.236	**0.292**	+23.6%
conceptual, sugya credit	0.063	0.133	**0.178**	+34.2%

Table 7: MRR on 552 held-out queries from Sanhedrin and Yevamot, retrieved over 81,481 segments. R@10 moves in step: conceptual questions 0.161 → 0.221 (+37.1%), practical 0.295 → 0.377 (+27.6%).

7.4 The pre-registered criterion

Before training we fixed the following: *the model must beat zero-shot on held-out question queries by a margin exceeding its gain on ruling queries.* A model improving only on rulings

has learned the benchmark rather than the task — and since a ruling shares vocabulary with its source sugya in a way a question does not, that is the easy win to optimise for accidentally.

credit	rulings	questions	verdict
strict	+12.2%	+27.9%	**pass**
sugya-level	+14.2%	+28.9%	**pass**

7.5 The gradient is the evidence

register	token overlap with target	MRR gain
ruling	0.41	+12%
practical question	0.30	+20%
conceptual question	0.21	+**36**%

Improvement rises monotonically as lexical overlap falls. The model gained most exactly where surface matching helps least — the signature of semantic learning, and a pattern a memorising model cannot produce, since memorisation would show the opposite gradient. Against BM25 the contrast is starker still: 2.3× on practical questions and 2.8× on conceptual ones, where BM25 reaches MRR 0.042 and is effectively unusable.

7.6 What this is not

Absolute performance is modest. Conceptual questions reach MRR 0.116 and R@1 0.067: the correct sugya ranks first roughly seven times in a hundred. This is a measurable improvement on a hard task, not a solved problem.

Three further caveats. The held-out tractates are heavy in status-chain material (levirate marriage, capital courts) that our question-writers independently reported as hardest to phrase naturally; we verified the split is not biased on living-law content (0.28 test vs 0.26 train), but the task is genuinely harder there. The test questions were generated from the rulings by the same process as the training questions, so the *relative* comparison is sound while the absolute level is an optimistic bound. And one epoch at batch 16 is a hardware compromise, not a considered choice.

8 Conclusion

The *ein mishpat* apparatus is a hand-curated cross-register relevance dataset of some fifty thousand pairs, assembled for exactly the retrieval task a modern system wants to perform. Fine-tuning on it produces a model that is better at the task users actually perform — asking questions — than at the benchmark it was trained against, by a factor of more than two.

Equally load-bearing is what measurement eliminated: dense-only retrieval (lexical and dense are complementary, union R@10 0.836 against $\approx$ 0.73 each); sugya-level indexing (a ten-point apparent gain that inverts under a pool-size control); ruling-anchored training (the benchmark task is not the deployed task); and exact-segment evaluation (it scores a capability the product does not provide, understating dense retrieval by roughly half). Each of the four would have looked correct on an uncontrolled measurement, and one of them looked correct on ours until the control was applied.

References

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